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[18MSCH1H01]

FIRST SEMESTER M.Sc. UNIVERSITY EXAMINATION, NOVEMBER 2019

INORGANIC CHEMISTRY - I

Time :03Hours **Maximum Marks:70**

Instructions:
1. **Question 1 is compulsory, carries 10 marks.**
2. **Questions 2 to 7 carry 15 marks each, answer any four.**

1. Answer any FIVE of the following questions: **5×2=10**

a. Write the stable oxidation state of Th and Pb, justify.

b. Arrange the following in the increasing order of their bond order;

O_2, N_2, NO, H_2 and He_2^+.

c. Write the structure of S_4N_4.

d. Silica possess tetrahedral geometry, justify.

e. AgCl readily dissolves in liq. NH_3, substantiate.

f. Define nuclear fission yield.

2. a. Explain the structure of $XeOF_4$, ClO_3F and SO_3 on the basis of VSEPR theory.

b. Using the Born-Haber cycle, calculate the lattice energy of CaO.
(Data: electron affinity of oxygen EA1=−141kJ/mol, EA2=744kJ/mol; ionization energy of calcium IE1=590kJ/mol, IE2=1145kJ/mol, the enthalpy of sublimation for calcium ΔHsub=178kJ/mol)

c. What is Fajan rule? Explain with an example. **6+4+5**

3. a. Sketch the molecular orbital energy level diagram for O_2^{-2} molecular ion.

b. Explain the formation of covalent bonds in SO_4^{-2} ion with the help of MOT.

c. Comment on the bond order and magnetic property of HCl molecule with a neat diagram of its molecular orbital energy level diagram. **5+5+5**

4. a. Write a note on the bonding and chemical properties of borazine.

b. What are phosphazenes? Comment on their structure and properties.

c. Explain the trend and anomalies of ionic radius and electronegativity along the period and down the group. **5+5+5**

5. a. Write notes on the following;

i. nesosilicates, **ii.** silica gel, and **iii.** zeolites.

b. Explain the following;

i. tetrahedral geometry of silicates, and **ii.** isomorphous replacement. **10+5**

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6. a. Describe the following;

i. Lewis acid-base, and ii. Bronsted-Lowry acid-base concepts.

b. Classify the following into acids or bases;

H_2O, NH_3, H_2SO_4, O^{2-}, H_2O, Mg^{2+}, NH_4^+, H_3O^+, SO_2 and Fe^{3+}.

c. Write a note on the hard and soft acid and base concept. **5+5+5**

7. a. Describe the evidences for nuclear shell model.

b. List the differences between nuclear forces and coulombic forces.

c. Write notes on the following;

i. nuclear chain reactions, and ii. radioactive carbon dating. **5+5+5**